Tom Derrick

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PROFILE

I'm a data scientist with four years across forecasting, mathematical optimisation, and applied AI — currently leading the development behind Modo Energy's Australian products. My work runs the full stack: MILP models solving over real transmission networks, ML price forecasters, the AWS pipelines that run them, and the React frontends customers touch. I direct our NEM model roadmap, reporting to the Head of Region, and train and delegate to a junior data scientist. I'm also the technical voice for the product in front of customers — on pre-sales calls, and presenting each release at customer workshops.

EXPERIENCE

Data Scientist — Modo Energy, Sydney Aug 2025 – present

Series B scaleup building AI-driven forecasting and investment intelligence for the energy transition — customers include Goldman Sachs, Apollo, EDF, and Shell. I joined the Australian team in its first year and lead the modelling behind its local products.

- Designed and shipped the nodal transmission network model behind the July 2026 forecast release — DC optimal power flow, bus-level dispatch, PTDF, load disaggregation, transmission augmentation through 2050+. Took the product from regional to 200+ nodes: largest update since the Australian launch and its first on-time release, delivered as sole modeller while training a new data scientist.
- Designed the capacity expansion model (linopy + XPRESS MILP) that replaced weeks of manual capacity-buildout tuning per forecast with a single optimisation at minimal compute cost.
- Build and run the production cost models producing long-term NEM price forecasts; battery dispatch optimisation and revenue modelling in the multi-market Global Dispatch Model (Pyomo + XPRESS).
- Built the full-stack contracting strategy system — backend valuation engine plus React configuration UX shipped to customers.
- Built and deployed an LLM support agent (Claude + RAG over model docs, code, and outputs) answering customer and internal questions across Slack and Intercom; contributed data engineering and semantic table annotation to Modo's AI assistant.
- Act as the forecasting expert on customer calls alongside the sales team — technical due diligence with prospects evaluating the product, and discovery with existing customers whose requirements feed back into the NEM model roadmap I direct.
- Present each forecast release at Modo's customer workshops and to the wider business — translating model and methodology changes into what they mean for investment and trading decisions.
- Author Modo's published forecast research — model releases, market outlooks, and deep dives.

Market Analyst (Graduate, Jul 2022 – Feb 2023) — Delta Electricity, Sydney Jul 2022 – Jul 2025

- Designed whole-of-market NEM simulation models (PLEXOS/Prophet) used for trading, hedging, and business development.
- Built an ML-powered spot-price forecasting model and a Monte-Carlo Gross-Margin-at-Risk engine for risk compliance and portfolio optimisation.
- Modelled major investment decisions — including the end-of-life extension of Vales Point Power Station — and LGC/ACCU certificate trading strategy.
- Optimised the planned-outage schedule to minimise revenue impact; built the internal live trading-performance dashboard. Completed Delta's 2024 future-leadership program.

Customer Success Manager — TechTarget, Sydney Nov 2020 – Jun 2022

- Managed APAC campaign delivery for marquee accounts (Red Hat, Dell, IBM, Cisco, Alibaba).

PERSONAL PROJECTS

- **gridline** — energy-market simulator built to stand in for PLEXOS, visual-first and agent-native: chronological LP dispatch (copperplate, zonal or nodal DC-OPF) with prices from duals and a “why this price” explain, storage, hydro water values and offer curves; builds the NEM from AEMO's ISP PLEXOS model. Python · linopy · HiGHS · React.
- **nem-market-watch** (nemdash.up.railway.app) — live dashboard for Australia's National Electricity Market: five-minute prices and demand, market metrics, PASA reserve outlook, bid-band analysis, and a tool-calling LLM agent that answers questions against live data. React · FastAPI · PostgreSQL · Railway.
- **nem-forward** (forward-frontend-production.up.railway.app) — XGBoost price forecaster (7-day, half-hourly, P10–P90) benchmarked against AEMO pre-dispatch, LP battery dispatch optimiser, and bid-band generation; every forecast persisted for later scoring against actuals. XGBoost · FastAPI · React.
- **nem-ingestion** — five-minute AEMO market data pipeline: six NEMWEB clients, sole writer and schema owner across 13 tables, backfill CLIs, hourly rollups, and a retention sweep that cut the database from 21 GB to 2 GB. Python · PostgreSQL · Railway.
- **pocket-etrm** — local-first ETRM phone app for a NEM trading desk: futures, bespoke OTC shapes and LGC/ACCU certificates, with positions, MtM, P&L, exposure, limits and parametric VaR marked against live ASX settlements. TypeScript · Expo · SQLite.
- **agent-tui** (github.com/tder311/agent-tui) — terminal UI for local AI coding agents: git worktrees, branches, Claude Code / OpenCode sessions, live running agents, and open PRs, grouped by project via app-agnostic filesystem discovery; read-only by default. Go · Bubble Tea.
- **opencode-orchestrator** (github.com/tder311/opencode-orchestrator · npm) — opencode plugin for parallel AI coding agents: each spawns into an isolated git worktree with its own interactive terminal session, pushing completion/blocker reports back to a supervising session.
- **railway-tui** (github.com/tder311/railway-tui) — terminal dashboard for Railway in Go and Bubble Tea: single static binary, OAuth (PKCE) login, live view of deployments, log streams, and environment variables over Railway's GraphQL API, with restart/redeploy/rollback actions and rate-limit-aware polling.
- **work-graph** — git-backed dependency tracker for work shared by people and AI agents: CLI, web and terminal UI over one Python core, with per-task git merges so two machines can edit at once. Python · Textual.
- **orchestrator** — coordinates multiple AI coding agents working in parallel across codebases: worktree isolation, task routing, session management.

EDUCATION

- UNSW — Master of Data Science (with Excellence)
- University of Sydney — BSc, Computer Science & Statistics
- AFMA Financial Markets Accreditation

SKILLS

Python · linopy · Pyomo · XPRESS · XGBoost · AWS (Batch, S3) · FastAPI · React · PostgreSQL · Claude API